

Parameter Configuration.

Dropout = 0.1
batch_size=32
SEED = 42
layers = 2
optimizer='adam'
loss='mse'
WINDOW=100
Units=32
1 Year Data
RMSE (Root Mean Squared Error)
MAE (Mean Absolute Error)

Layer (type)	Output Shape	Param #
lstm (LSTM)	(None, 100, 32)	4,352
dropout (Dropout)	(None, 100, 32)	0
lstm_1 (LSTM)	(None, 32)	8,320
dropout_1 (Dropout)	(None, 32)	0
dense (Dense)	(None, 1)	33
Total params: 12,705 (49.63 KB)		
Trainable params: 12,705 (49.63 KB)		
Non-trainable params: 0 (0.00 B)		

Epochs (5) Patience (3)

```
Epoch 1/5
1397/1397 ————— 104s 72ms/step - loss:
0.0116 - val_loss: 1.9982e-04
Epoch 2/5
1397/1397 ————— 139s 69ms/step - loss:
0.0012 - val_loss: 1.5432e-04
Epoch 3/5
1397/1397 ————— 97s 69ms/step - loss:
7.6997e-04 - val_loss: 1.3901e-04
Epoch 4/5
1397/1397 ————— 99s 71ms/step - loss:
4.7728e-04 - val_loss: 1.2989e-04
Epoch 5/5
1397/1397 ————— 98s 70ms/step - loss:
3.3230e-04 - val_loss: 1.2299e-04

1397/1397 ————— 28s 20ms/step - loss:
2.8724e-05
MSE Train: 3.9842689147917554e-05
464/464 ————— 9s 19ms/step - loss:
1.0584e-04
MSE Test: 0.00012298976071178913
464/464 ————— 8s 18ms/step - loss:
9.6633e-05
MSE Val: 0.00010950281284749508

Test Set RMSE (Original Scale): 0.000926 EURUSD
Test Set MAE (Original Scale): 0.000645 EURUSD
Validation Set RMSE (Original Scale): 0.000873 EURUSD
Validation Set MAE (Original Scale): 0.000565 EURUSD
Training Set RMSE (Original Scale): 0.000527 EURUSD
Training Set MAE (Original Scale): 0.000361 EURUSD
```

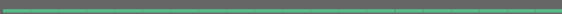
Epochs (10) Patience (5)

```
Epoch 1/10
1397/1397 ██████████ 107s 73ms/step - loss:
2.8873e-04 - val_loss: 1.0659e-04
Epoch 2/10
1397/1397 ██████████ 149s 78ms/step - loss:
2.7145e-04 - val_loss: 1.3248e-04
Epoch 3/10
1397/1397 ██████████ 135s 73ms/step - loss:
2.6579e-04 - val_loss: 9.2855e-05
Epoch 4/10
1397/1397 ██████████ 142s 73ms/step - loss:
2.7631e-04 - val_loss: 1.2890e-04
Epoch 5/10
1397/1397 ██████████ 100s 72ms/step - loss:
2.5894e-04 - val_loss: 1.0389e-04
Epoch 6/10
1397/1397 ██████████ 147s 76ms/step - loss:
2.5830e-04 - val_loss: 9.1989e-05
Epoch 7/10
1397/1397 ██████████ 102s 73ms/step - loss:
2.5801e-04 - val_loss: 1.0073e-04
Epoch 8/10
1397/1397 ██████████ 142s 73ms/step - loss:
2.6012e-04 - val_loss: 1.2195e-04
Epoch 9/10
1397/1397 ██████████ 140s 72ms/step - loss:
2.5354e-04 - val_loss: 1.5391e-04
Epoch 10/10
1397/1397 ██████████ 102s 73ms/step - loss:
2.4842e-04 - val_loss: 8.6664e-05

1397/1397 ██████████ 28s 20ms/step - loss:
1.9602e-05
MSE Train: 3.2247731724055484e-05
464/464 ██████████ 9s 20ms/step - loss:
8.8152e-05
MSE Test: 8.666364738019183e-05
464/464 ██████████ 9s 19ms/step - loss:
7.7052e-05
MSE Val: 8.038169471547008e-05
```


Epochs (15) Patience (7)

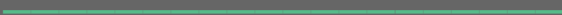
Epoch 1/15

1397/1397  **82s** 56ms/step - loss: 0.0147 - val_loss: 2.0921e-04

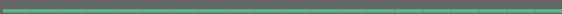
Epoch 2/15

1397/1397  **79s** 57ms/step - loss: 0.0013 - val_loss: 1.9619e-04

Epoch 3/15

1397/1397  **76s** 54ms/step - loss: 8.1578e-04 - val_loss: 1.6421e-04

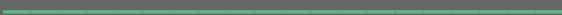
Epoch 4/15

1397/1397  **75s** 54ms/step - loss: 5.1583e-04 - val_loss: 1.4289e-04

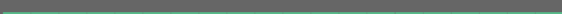
Epoch 5/15

1397/1397  **78s** 56ms/step - loss: 3.4279e-04 - val_loss: 1.9670e-04

Epoch 6/15

1397/1397  **81s** 58ms/step - loss: 2.9466e-04 - val_loss: 2.2567e-04

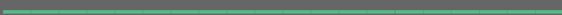
Epoch 7/15

1397/1397  **79s** 57ms/step - loss: 2.7616e-04 - val_loss: 1.3590e-04

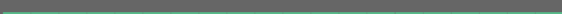
Epoch 8/15

1397/1397  **83s** 59ms/step - loss: 2.6512e-04 - val_loss: 1.0527e-04

Epoch 9/15

1397/1397  **79s** 56ms/step - loss: 2.6884e-04 - val_loss: 1.4249e-04

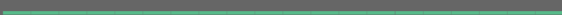
Epoch 10/15

1397/1397  **80s** 57ms/step - loss: 2.6231e-04 - val_loss: 8.8442e-05

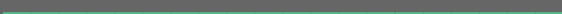
Epoch 11/15

1397/1397  **79s** 56ms/step - loss: 2.5739e-04 - val_loss: 1.1936e-04

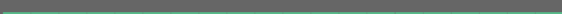
Epoch 12/15

1397/1397  **76s** 54ms/step - loss: 2.6369e-04 - val_loss: 1.1569e-04

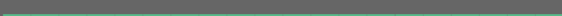
Epoch 13/15

1397/1397  **78s** 55ms/step - loss: 2.5861e-04 - val_loss: 1.2615e-04

Epoch 14/15

1397/1397  **82s** 59ms/step - loss: 2.5626e-04 - val_loss: 6.4743e-05

Epoch 15/15

1397/1397  **80s** 57ms/step - loss: 2.5163e-04 - val_loss: 1.1186e-04

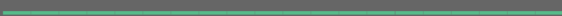
1397/1397  **20s** 14ms/step - loss: 4.2312e-05

Epochs (20) Patience (10)

```
Epoch 1/20
1397/1397 ————— 106s 73ms/step - loss:
2.4294e-04 - val_loss: 1.4577e-04
Epoch 2/20
1397/1397 ————— 140s 72ms/step - loss:
2.3240e-04 - val_loss: 5.0715e-05
Epoch 3/20
1397/1397 ————— 102s 73ms/step - loss:
2.3937e-04 - val_loss: 1.6397e-04
Epoch 4/20
1397/1397 ————— 139s 71ms/step - loss:
2.4675e-04 - val_loss: 6.0913e-05
Epoch 5/20
1397/1397 ————— 142s 71ms/step - loss:
2.3612e-04 - val_loss: 1.8130e-04
Epoch 6/20
1397/1397 ————— 142s 71ms/step - loss:
2.3865e-04 - val_loss: 5.6910e-05
Epoch 7/20
1397/1397 ————— 144s 72ms/step - loss:
2.3683e-04 - val_loss: 8.1295e-05
Epoch 8/20
1397/1397 ————— 140s 71ms/step - loss:
2.3963e-04 - val_loss: 5.1375e-05
Epoch 9/20
1397/1397 ————— 99s 71ms/step - loss:
2.3267e-04 - val_loss: 7.5308e-05
Epoch 10/20
1397/1397 ————— 102s 73ms/step - loss:
2.3996e-04 - val_loss: 4.8875e-05
Epoch 11/20
1397/1397 ————— 142s 73ms/step - loss:
2.3371e-04 - val_loss: 4.2777e-05
Epoch 12/20
1397/1397 ————— 101s 72ms/step - loss:
2.3058e-04 - val_loss: 1.0120e-04
Epoch 13/20
1397/1397 ————— 148s 76ms/step - loss:
2.2951e-04 - val_loss: 5.1202e-05
Epoch 14/20
1397/1397 ————— 136s 72ms/step - loss:
2.2806e-04 - val_loss: 1.5814e-04
Epoch 15/20
1397/1397 ————— 100s 72ms/step - loss:
2.3452e-04 - val_loss: 4.2113e-05
Epoch 16/20
```


Epochs (20) Patience (13)

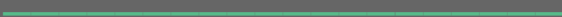
Epoch 1/20

1397/1397  **81s** 56ms/step - loss: 2.5570e-04 - val_loss: 6.2413e-05

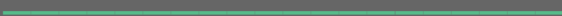
Epoch 2/20

1397/1397  **74s** 53ms/step - loss: 2.5841e-04 - val_loss: 1.0052e-04

Epoch 3/20

1397/1397  **80s** 57ms/step - loss: 2.4632e-04 - val_loss: 8.8310e-05

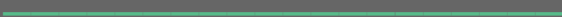
Epoch 4/20

1397/1397  **78s** 56ms/step - loss: 2.5290e-04 - val_loss: 6.3833e-05

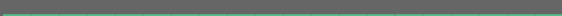
Epoch 5/20

1397/1397  **71s** 51ms/step - loss: 2.3462e-04 - val_loss: 8.4940e-05

Epoch 6/20

1397/1397  **80s** 57ms/step - loss: 2.5271e-04 - val_loss: 1.1515e-04

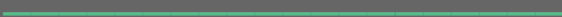
Epoch 7/20

1397/1397  **76s** 54ms/step - loss: 2.4691e-04 - val_loss: 5.7238e-05

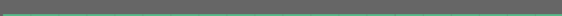
Epoch 8/20

1397/1397  **76s** 54ms/step - loss: 2.4062e-04 - val_loss: 1.3160e-04

Epoch 9/20

1397/1397  **73s** 52ms/step - loss: 2.4773e-04 - val_loss: 1.0341e-04

Epoch 10/20

1397/1397  **76s** 55ms/step - loss: 2.3681e-04 - val_loss: 6.1487e-05

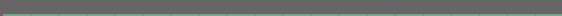
Epoch 11/20

1397/1397  **79s** 57ms/step - loss: 2.4094e-04 - val_loss: 5.1818e-05

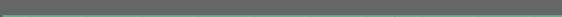
Epoch 12/20

1397/1397  **81s** 58ms/step - loss: 2.4068e-04 - val_loss: 4.7754e-05

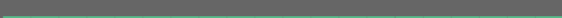
Epoch 13/20

1397/1397  **74s** 53ms/step - loss: 2.4515e-04 - val_loss: 5.3554e-05

Epoch 14/20

1397/1397  **74s** 53ms/step - loss: 2.3204e-04 - val_loss: 7.0733e-05

Epoch 15/20

1397/1397  **74s** 53ms/step - loss: 2.4165e-04 - val_loss: 6.9489e-05

Epoch 16/20

1397/1397  **80s** 57ms/step - loss: 2.3547e-04 - val_loss: 1.2269e-04

